

Supplemental Detail (Expanded Comment from DSI)

To: Department of Health and Human Services (HHS)
Assistant Secretary for Technology Policy (ASTP) and Office of the National Coordinator for Health Information Technology (ONC)

Submitted via: Regulations.gov

Re: "HHS Health Sector AI RFI" (RIN 0955-AA13)

Introduction

On behalf of the Dental Standards Institute (DSI), we appreciate the opportunity to respond to this Request for Information (RFI) regarding the acceleration of artificial intelligence (AI) adoption and use as part of clinical care (2025-23641).

DSI is a standards development organization focused on emerging technology, interoperability, and patient-centered care in dentistry. Historically, dentistry lacked a unified, technically focused standards authority aligned with modern health IT and interoperability priorities. DSI was formed to fill this gap and has already made substantial progress in advancing the foundational standards required for dental interoperability, patient-centered data portability, and responsible technology adoption.

Information about DSI is available at dentalstandardsinstitute.com.

HHS has a unique opportunity to ensure dentistry is not excluded from national AI policy and infrastructure development. Dentistry is one of the largest, most frequently accessed clinical domains in the United States, yet it remains structurally disconnected from the broader healthcare data ecosystem. AI adoption will not succeed at scale without trust infrastructure, including interoperability, standardized data capture, independent validation, and clear accountability structures.

Overarching Position

AI can improve health outcomes, reduce administrative burden, and expand access to care. However, accelerated AI adoption without standards and transparency risks producing a

fragmented ecosystem of opaque tools, inconsistent performance claims, and avoidable patient harm.

Dentistry is particularly sensitive to these risks because:

- diagnostic coding is inconsistently used
- imaging workflows vary widely by practice and device
- clinical documentation remains largely unstructured
- dental and medical records remain disconnected
- vendor validation is often self-reported and difficult to compare across tools

For dentistry to contribute meaningfully to national AI objectives, HHS must prioritize policies that reinforce human oversight, independent validation, and interoperability, while aligning reimbursement structures to support diagnosis-based care.

Key Recommendations to HHS

1. Clinical AI Must Be Decision-Support, Not Decision-Substitution

DSI strongly urges HHS to reinforce that AI used in clinical care must remain decision-support, not decision-substitution.

Dentistry is not ready for autonomous clinical AI. Even where AI tools demonstrate strong performance in controlled settings, real-world deployment introduces risks including imaging variation, bias, drift, and workflow overreliance. The most dangerous failure mode is silent normalization, where AI output becomes treated as clinical truth without sufficient human review.

HHS should establish clear expectations that:

- human review remains mandatory for AI outputs affecting diagnosis, treatment planning, or documentation
- AI-generated recommendations must be traceable, auditable, and version-controlled
- accountability remains clear: clinicians and organizations remain responsible for clinical decisions
- post-deployment monitoring must be expected, not optional

Accelerating adoption without reinforcing human oversight will erode public trust and increase preventable patient harm.

2. Dentistry's Treatment-Based Structure Limits AI: Diagnosis Must Become Standard and Reimbursable

Dentistry has historically been treatment-driven rather than diagnosis-driven. Unlike medicine, dentistry has not consistently implemented diagnostic coding in routine practice, and the reimbursement ecosystem has not strongly incentivized diagnosis capture.

This structural issue is a major barrier to safe, scalable AI adoption.

AI systems depend on structured diagnostic labeling to train, validate, and improve. When a clinical domain operates primarily through procedure documentation, models risk being optimized for predicting treatment patterns rather than reflecting clinical truth and patient outcomes.

Additionally, the lack of standardized diagnosis capture limits dentistry's ability to participate in:

- integrated oral-systemic care
- longitudinal chronic disease management
- value-based care models
- research and real-world evidence generation
- cross-disciplinary AI decision support

DSI recommends that HHS, including CMS, prioritize policies that accelerate diagnosis-based dentistry by supporting:

- routine diagnostic coding in dental workflows
- adoption of standardized diagnostic terminologies
- reimbursement models that recognize diagnosis as a reimbursable clinical act

If dentistry is expected to contribute to national AI-enabled clinical care goals, it must be structurally enabled to document diagnosis in a consistent, interoperable manner.

3. Independent External Validation Must Become a Norm for Dental AI Tools

A major barrier to trust and adoption of dental AI tools is the absence of consistent independent validation.

Most dental AI products today rely on:

- proprietary datasets
- internal testing protocols
- limited transparency into training populations
- non-standard reporting of sensitivity/specificity
- inconsistent disclosure of subgroup performance
- unclear post-deployment drift monitoring

Self-validation is not validation.

HHS should prioritize efforts that support the creation and scaling of independent benchmarking and validation frameworks for AI tools used in clinical care, including dentistry. Such efforts could include:

- third-party audit mechanisms
- standardized evaluation metrics and reporting requirements
- independent testing datasets representing diverse populations and real-world imaging variability
- mandatory subgroup performance reporting
- post-deployment monitoring expectations and drift detection requirements

This aligns directly with HHS's emphasis on evaluation methods, robustness testing, and governance structures that promote trust (2025-23641).

4. Interoperability is the Primary Bottleneck to High-Value AI in Dentistry

HHS asks directly where enhanced interoperability would widen market opportunities and accelerate AI development (2025-23641). Dentistry is a clear answer.

The most valuable AI use cases in dentistry, those that improve systemic outcomes and reduce total cost of care, depend on longitudinal data exchange between dental and medical systems. However, dental data remains highly fragmented across practice management systems, imaging systems, payer workflows, and unstructured clinical notes.

Without interoperability, AI will remain limited to isolated point solutions rather than contributing to true healthcare transformation.

DSI recommends HHS prioritize interoperability policies that explicitly include dentistry, including support for:

- structured oral health data exchange standards
- dental imaging interoperability and metadata normalization
- consistent capture and exchange of diagnostic findings
- alignment between dental standards and broader medical exchange frameworks (including FHIR-based infrastructure)
- patient-centered portability of dental records

DSI has already advanced work in dental interoperability and patient-centered data portability, and we encourage HHS to support and accelerate national adoption of these standards-based approaches.

5. HHS Should Support a National Dental AI Standards and Validation Partnership

Dentistry historically lacked a unified infrastructure for AI governance, interoperability alignment, and standardized evaluation. DSI was formed to address this gap and has already begun advancing the foundational work required to support responsible AI adoption in dentistry.

We recommend HHS/ONC support and accelerate the development of a national dental AI trust framework, including:

- standardized validation datasets across multiple imaging devices and clinical settings
- third-party testing and certification mechanisms
- minimum transparency requirements for dental AI vendors
- standardized performance reporting formats
- audit log requirements for AI recommendations in clinical workflows
- guidance for post-deployment monitoring and model drift surveillance

This framework should support private sector innovation while ensuring consistent trust, accountability, and patient safety.

DSI stands ready to convene clinicians, DSOs, vendors, payers, and patient advocates to support HHS in defining and operationalizing this infrastructure.

6. Standardized Transparency Disclosures (“AI Nutrition Labels”) Should Be Encouraged

A practical and high-impact policy step would be to encourage standardized disclosures for AI tools used in clinical care.

DSI recommends that HHS promote a consistent “AI nutrition label” disclosure format including:

- intended use and clinical scope
- dataset provenance and size
- demographic distribution of training/testing sets
- performance metrics by subgroup
- known failure conditions and limitations
- model update frequency and version control approach
- drift monitoring approach
- documentation requirements for human review

Such disclosures would reduce uncertainty for purchasers, support responsible adoption, and increase public trust.

7. Dentistry is a High-Leverage Domain for Prevention and Oral-Systemic Health Integration

Dentistry represents a major opportunity to strengthen national prevention strategy.

Oral inflammation and dental disease correlate strongly with broader chronic disease risk. Dental visits are often more frequent than physician visits for many Americans, meaning dentistry is a critical clinical setting for early risk detection and prevention workflows.

If AI is responsibly implemented, with interoperability, diagnostic modernization, and independent validation, dentistry can serve as a high-leverage entry point for improving systemic health outcomes while reducing downstream healthcare costs.

Responses Aligned to HHS Specific Questions

The recommendations above align directly with multiple questions posed in the RFI (2025-23641).

Question 1: Biggest barriers to innovation and adoption

- limited interoperability of dental data
- lack of consistent diagnostic coding adoption
- inconsistent imaging workflows and device variability
- opaque vendor validation and lack of independent benchmarking
- unclear accountability and auditability expectations

Question 2: Regulatory, payment policy, and programmatic changes

- accelerate diagnostic coding adoption in dentistry
- modernize reimbursement incentives to support diagnosis capture
- expand interoperability priorities to include dentistry explicitly
- establish clear expectations for auditability and human oversight

Question 3: Governance and accountability issues

- clinicians remain accountable for clinical decisions
- AI recommendations must be logged and auditable
- liability and indemnification expectations must remain clear

Question 4: Evaluation methods and robustness testing

- independent benchmarking datasets
- standardized subgroup reporting
- real-world multi-site testing
- post-deployment performance monitoring and drift detection

Question 5: Accreditation, certification, and testing

- support third-party testing ecosystems
- encourage industry-wide certification models

- tie incentives to independently validated performance, not marketing claims

Question 8: Interoperability opportunities

- structured oral health data exchange
- dental imaging interoperability and metadata standards
- diagnostic representation and portability
- alignment with broader medical interoperability frameworks

Conclusion

HHS has a timely opportunity to accelerate AI adoption in a way that improves outcomes, reduces burden, and strengthens public trust. Dentistry must be explicitly included in this effort.

AI adoption will not scale safely without trust infrastructure: interoperability standards, diagnostic modernization, independent validation, and clear accountability. Dentistry historically lacked unified leadership in these areas. DSI was formed to fill this gap and has already advanced foundational work in interoperability, patient-centered record portability, and calibration efforts that support consistent clinical and payer-facing workflows.

DSI urges HHS to prioritize:

1. mandatory human oversight and accountability
2. modernization of diagnosis capture and reimbursement alignment in dentistry
3. independent external validation frameworks for dental AI tools
4. dental interoperability as a prerequisite for scalable clinical AI
5. standardized transparency disclosures for AI tools
6. real-world monitoring expectations for deployed AI systems

DSI welcomes the opportunity to collaborate with HHS/ONC, CMS, NIH, and FDA to accelerate responsible AI adoption in dentistry and to ensure dentistry becomes an integrated contributor to national health outcomes.

Respectfully submitted,
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